

ZHIYU SUN

zs2710@columbia.edu | 646-552-5444 | zhiyusun.com

SKILLS

- Languages: Python, C++, MATLAB, C, TypeScript, Bash, SQL, C#, R, LaTeX
- Libraries: NumPy, TensorFlow, Pandas, Matplotlib, Qt, PsychoPy
- Tools: Git, SolidWorks, Simulink, VS Code, Bambu Studio, Power BI

EXPERIENCE

VisionNav Robotics, Inc.

Atlanta, GA

Project Engineer

Apr 2026 – Sep 2026

- Handled on-site commissioning independently, coordinating remotely with 3 cross-functional teams to resolve technical issues and translating customer requirements into technical solutions.
- Delivered 3 AGV automation solutions for leading global manufacturers, incorporating Ethernet configuration, 3D mapping, route planning, task scheduling, PLC integration, and customer training.
- Calibrated AGV fork positioning, steering, and LiDAR sensors, validated point cloud alignment and configured laser safety zones for precise cargo handling, autonomous navigation, and obstacle avoidance.

SurgeVance, Inc.

New York, NY

Data Science R&D Intern

Oct 2025 – Apr 2026

- Built a TypeScript image annotation tool using OpenSeadragon and SVG overlays, applying coordinate transformations and event-driven synchronization to maintain consistent annotations across image views.
- Reduced segmentation time by 60% through parallel heatmap loading and saving with ThreadPoolExecutor and GPU-accelerated segmentation mask computation with CPU/Numba fallbacks.

Columbia University

New York, NY

Research Assistant

Feb 2025 – Jun 2025

- Developed a multimodal model integrating volumetric MRI and plasma biomarkers for Alzheimer's disease assessment, achieving 88.4% disease-stage classification accuracy.
- Optimized a CNN-Mamba model for 3D MRI segmentation through CPU-side data preloading and streamlined Dice computation, reducing training time by 75% while maintaining a Dice score of 0.988.

Shanghai Microport Medbot (Group) Co., Ltd.

Shanghai, CN

Operations Intern

Jun 2023 – Sep 2023

- Demonstrated robotic technologies to 50+ visitors daily across 4 exhibitions, securing 13 hospital-director contacts and translating customer feedback and clinical operation data into 2 team-adopted improvement plans.
- Developed a hierarchical, parameter-based data-sorting model to streamline inventory tracking and reduced inventory processing time by 50%.

PROJECTS

AI-Enabled Quadruped Robot with Vision and Audio Control

New York, NY

- Built a quadruped robot with custom 3D-printed leg assemblies, integrating a Raspberry Pi 4 with a servo controller and daisy-chained bus servos for joint actuation.
- Integrated Gemini and YOLOv5n for visual perception, Porcupine for wake-word detection, and Vosk for speech recognition to enable real-time voice control.

Cognitive Task Design in Python

New York, NY

- Designed a memory task presenting image stimuli and recording behavioral responses using PsychoPy.
- Integrated the task and real-time EEG recording into LSL for synchronized data acquisition.

Automated Financial Reporting System

New York, NY

- Built a financial reporting application integrating Power BI reports through REST APIs and automating Excel data ingestion, filtering, and table generation using Microsoft Access.
- Integrated DeepSeek-V3 to analyze filtered financial data and automatically generate PowerPoint reports.

EDUCATION

Columbia University

New York, NY

Master of Science in Biomedical Engineering

Sep 2024 – Dec 2025